Homework Solutions

Linear Algebra

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Problem Q1

What does it mean for a square matrix to be invertible? State a condition on the determinant.

Unsolved: Conceptual question requires LLM solver (student extension point).

Problem Q2

What is an eigenvalue of a matrix A? Write the defining equation.

Unsolved: Conceptual question requires LLM solver (student extension point).

Problem P1

Compute the determinant of the matrix

$$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$$

Solution. $\det \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = -2$

determinant $= -2$

Answer:

$$\boxed{-2}$$

Problem P2

Find the inverse of the matrix

$$\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

Solution. $\left(\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \right)^{-1} = \begin{bmatrix} \frac{3}{5} & -\frac{1}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{bmatrix}$

Answer:

$$\boxed{\begin{bmatrix} \frac{3}{5} & -\frac{1}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{bmatrix}}$$

Problem P3

Find the eigenvalues and corresponding eigenvectors of the matrix

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Solution. eigenvalue $\lambda = 1$ (multiplicity 1)
corresponding eigenvector $= \begin{bmatrix} -1 \\ 1 \end{bmatrix}$
eigenvalue $\lambda = 3$ (multiplicity 1)
corresponding eigenvector $= \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
Answer:

$$\boxed{\lambda = 3, \quad \lambda = 1}$$

Problem P4

Solve the linear system $A\mathbf{x} = \mathbf{b}$ where

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

Solution. solution vector $= \left(\frac{8}{5}, \frac{9}{5}\right)$
Answer:

$$\boxed{\left\{\left(\frac{8}{5}, \frac{9}{5}\right)\right\}}$$

Problem P5

Compute the dot product of the vectors

$$\mathbf{u} = \langle 1, 2, 3 \rangle, \qquad \mathbf{v} = \langle 4, 5, 6 \rangle$$

Solution. dot product $= 32$
Answer:

$$\boxed{32}$$

Problem P6

Compute the cross product of the vectors

$$\mathbf{u} = \langle 1, 0, 0 \rangle, \qquad \mathbf{v} = \langle 0, 1, 0 \rangle$$

Solution. $u \times v = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

Answer:

$$\boxed{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}$$

Problem P7

Find the transpose of the matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Solution. $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$

Answer:

$$\boxed{\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}}$$

Problem P8

Multiply the matrices

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

Solution. $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$

Answer:

$$\boxed{\begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}}$$

Problem P9

Find the rank of the matrix

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Solution. $\operatorname{rank}\begin{bmatrix}1 & 2\\ 2 & 4\end{bmatrix} = 1$

Answer:

$$\boxed{1}$$

Problem P10

Find the trace of the matrix

$$\begin{bmatrix}2 & 1\\ 1 & 3\end{bmatrix}$$

Solution. $\operatorname{tr}\begin{bmatrix}2 & 1\\ 1 & 3\end{bmatrix} = 5$

Answer:

$$\boxed{5}$$

Auto-generated: 10/12 problems solved by SymPy.